



GIRRAWEE HIGH SCHOOL

TASK 1 2019 (December 2018)

MATHEMATICS

EXTENSION 2

Complex Numbers

Time allowed – 90 Minutes

DIRECTIONS TO CANDIDATES

- Attempt all questions.
- For Questions 1 -5, shade the circle for the letter corresponding to the correct answer on your answer sheet.
- For Questions 6-11, start each question on a new page. Each question should be clearly labelled.
- All necessary working must be shown for Questions 11– 15.
- Marks may be deducted for careless or badly arranged work.
- Board approved calculators may be used.
- A Mathematics reference sheet is provided.
- All diagrams are NOT TO SCALE.
- The use of the notation $\text{cis}\theta$ is acceptable in this examination.

Question 1

$$(-i)^{2019} =$$

(A) i

(B) $-i$

(C) 1

(D) -1

Question 2

$$\frac{\bar{z}}{|z|} =$$

(A) $\frac{1}{z}$

(B) $\frac{|z|}{z}$

(C) $\frac{z}{\bar{z}}$

(D) $\frac{1}{z|z|}$

Examination continues on the following page

Question 3

If $w = 2 + i$, $\frac{1}{w^2} =$

- (A) $\frac{3+4i}{25}$ (B) $\frac{3-4i}{25}$ (C) $\frac{3+4i}{5}$ (D) $\frac{3-4i}{5}$

Question 4

Multiplying by a complex number rotates the result $\frac{\pi}{6}$ CLOCKWISE about the origin. The complex number I multiplied by could be

- (A) $\frac{1-i\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}-i}{2}$ (C) $\frac{1+i\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}+i}{2}$

Question 5

When expressed in modulus/argument form to the nearest degree, $-4 - i =$

- (A) $\sqrt{17} (\cos 14^\circ + i \sin 14^\circ)$ (B) $\sqrt{17} (\cos 166^\circ + i \sin 166^\circ)$
 (C) $\sqrt{17} (\cos -166^\circ + i \sin -166^\circ)$ (D) $\sqrt{17} (\cos -14^\circ + i \sin -14^\circ)$

Question 6 (16 marks – show all workings in your answer booklet)**Marks**

- (a) (i) By realising the denominator, find the value $\frac{\sqrt{3}-i}{1+i}$ in Cartesian form. **2**
- (ii) By converting both $\sqrt{3} - i$ and $1 + i$ to modulus/ argument form, express **5**
 the value of $\frac{\sqrt{3}-i}{1+i}$ in modulus/argument form.
- (iii) Hence use your answers to (i) and (ii) to find the exact value of $\cos \frac{5\pi}{12}$ **1**
- (b) (i) By letting $(x + iy)^2 = 24 - 10i$ x, y real, find $\sqrt{24 - 10i}$ in Cartesian form. **3**
- (ii) Hence solve $z^2 + (1 + i)z - 6 + 3i = 0$. **2**
- (c) Find all five fifth roots of $16 - 16i\sqrt{3}$. Leave your answers in **3**
 modulus/argument form.

Examination continues on the following page

Question 7 (8 marks – show all workings in your answer booklet) **Marks**

(a) Sketch these loci on separate Argand diagrams if $z = x + iy$

(i) $|z + 3 - 4i| = 5$ **2**

(ii) $\text{Arg}(z + 3 - i) = \frac{2\pi}{3}$. **2**

(b) Shade the region in the complex plane where **4**

$$|z - 1 - i| \leq 5 \text{ and } \frac{\pi}{4} < \text{Arg}z \leq \frac{\pi}{2}$$

Question 8 (9 marks – show all workings in your answer booklet)

If $z = \cos\theta + i\sin\theta$

(i) Use DeMoivre's theorem to show that $z^n - z^{-n} = 2i\sin n\theta$. **2**

(ii) Hence express $\sin^5\theta$ in the form $A\sin 5\theta + B\sin 3\theta + C\sin\theta$. **3**

(iii) By substituting $\theta = \frac{\pi}{5}$ and using $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ show **4**

that $\sin \frac{\pi}{5}$ is a root of the equation $16x^4 - 20x^2 + 5 = 0$ and find

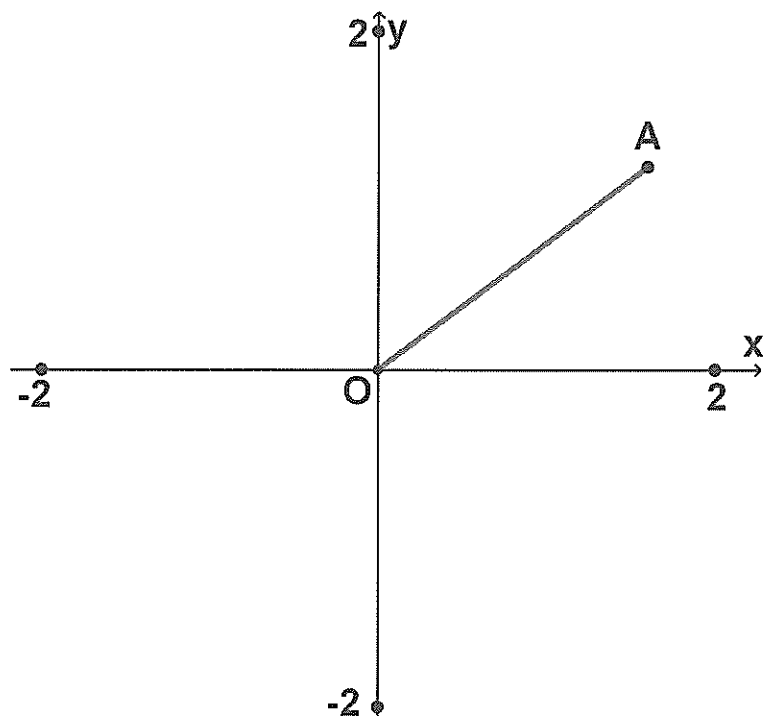
the exact value of $\sin \frac{\pi}{5}$. (You may leave your answer under a square root sign).

Examination continues on the following page

Question 9 (10 Marks – show all workings in your answer booklet)

Marks

(a) In the diagram below, $\overrightarrow{OA} = z$ where $|z| = 2$ and $0 < \text{Arg } z < \frac{\pi}{2}$.



Copy the diagram in to your answer booklet and draw on it the complex number

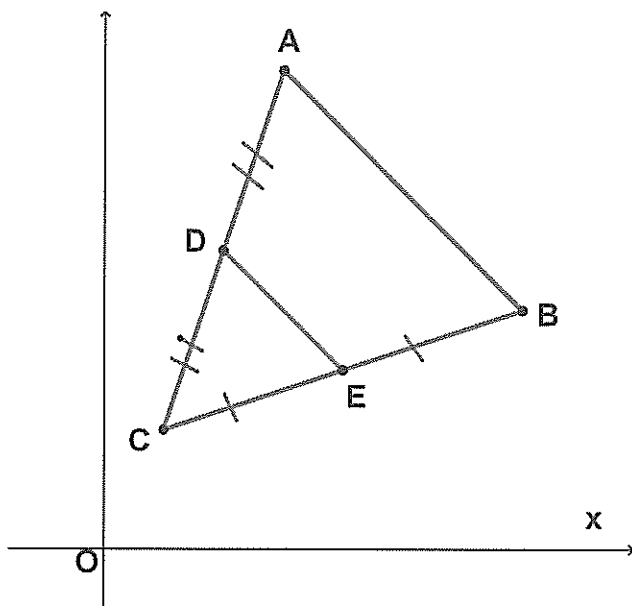
- | | |
|---|---|
| (i) $\overrightarrow{OB}$ so that $\overrightarrow{OB} = \bar{z}$ | 1 |
| (ii) $\overrightarrow{OC}$ so that $\overrightarrow{OC} = iz$ | 1 |
| (iii) $\overrightarrow{OD}$ so that $\overrightarrow{OD} = z(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3}))$ | 1 |
| (iv) $\overrightarrow{OE}$ so that $\overrightarrow{OE} = z + 2$ | 1 |
| (v) $\overrightarrow{OF}$ so that $\overrightarrow{OF} = z + i$ | 1 |

Question 9 continues on the following page

Question 9 (continued)

Marks

(b) In the Argand diagram below, A, B and C are the vertices of a triangle. D and E are the midpoints of AC and BC respectively.



If the complex number $\overrightarrow{OA} = a$, the complex number $\overrightarrow{OB} = b$ and $\overrightarrow{OC} = c$,

- (i) Find expressions for $\overrightarrow{DC}$ and $\overrightarrow{EC}$ in terms of a, b and c . 2
- (ii) HENCE show that $\overrightarrow{AB} = 2 \times \overrightarrow{DE}$. 3

Question 10 (10 Marks – show all workings in your answer booklet)

(a) Find the locus of z in terms of x and y if $z = x + iy$ and

- (i) $\frac{z-2i}{z+4}$ is entirely real. 3
- (ii) $\frac{|z-7-6i|}{|z+5+3i|} = 2$. 3

(You do not need to draw diagrams for either question in Part (a))

- (b) The locus $\text{Arg} \left\{ \frac{z-7-2i}{z+1-2i} \right\} = \frac{\pi}{3}$ represents part of a circle. 4

Draw this circle part on an Argand diagram and find its centre, radius and equation.

Examination continues on the following page

Question 11 (15 Marks – show all workings in your answer booklet) **Marks**

(a)(i) Solve $z^9 - 1 = 0$. 2

If w is the root of $z^9 - 1 = 0$ with the smallest positive argument

(ii) Show that $w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = -1$ 1

(You do NOT need to prove that w^2, w^3 etc. are the other non real roots of $z^9 - 1 = 0$)

(iii) Show that $w^8 = \frac{1}{w}$ 1

(iv) Show that $w + w^2 + w^3 + w^4 + \frac{1}{w} + \frac{1}{w^2} + \frac{1}{w^3} + \frac{1}{w^4} = -1$ 1

(v) Show that $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$ 2

(You may assume that $\cos n\theta + \cos(-n\theta) = 2\cos n\theta$).

(b) (i) Resolve $z^9 - 1$ into real linear and quadratic factors. Hence show that 4

$$z^6 + z^3 + 1 = (z^2 - 2z\cos \frac{2\pi}{9} + 1)(z^2 - 2z\cos \frac{4\pi}{9} + 1)(z^2 + 2z\cos \frac{\pi}{9} + 1)$$

(ii) Hence show that 2

$$2\cos 3\theta + 1 = (2\cos \theta - 2\cos \frac{2\pi}{9})(2\cos \theta - 2\cos \frac{4\pi}{9})(2\cos \theta + 2\cos \frac{\pi}{9})$$

(iii) Hence by an appropriate substitution for θ , show that 2

$$\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}.$$

END OF EXAMINATION!!!

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Complex Numbers (Task 1)
December 2018 for 2019 HSC

Multiple Choice

Q(1) A (2) B (3) B (4) B (5) C

(1) $(-i)^{2019} = (-i)^3 = i$ (A)

(2) $\frac{z}{|z|^2} = \frac{1}{z}$ (B)
 $\frac{z}{|z|^2} = \frac{1}{z} \Rightarrow \frac{z}{|z|} = \frac{1}{|z|}$

(3) $\frac{1}{w^2} = \frac{1}{(2+i)^2} = \frac{1 \times (3-i)}{(3+i)(3-i)} = \frac{3-i}{10}$ (B)

(4) Arg $= \frac{\pi}{6}$
 $= \frac{\sqrt{3}-i}{2}$ (B)

(5) $\Rightarrow -4-i = \sqrt{17} \operatorname{cis}(-166^\circ)$ (C)

6(a)(i) $\frac{\sqrt{3}-i}{1+i} \times \frac{(1-i)}{(1-i)} = \frac{(\sqrt{3}-1) - i(1+\sqrt{3})}{2}$

(ii) $\frac{\sqrt{3}-i}{1+i} = 2 \operatorname{cis}(-\frac{\pi}{6})$
 $1+i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$
 $\frac{\sqrt{3}-i}{1+i} = \frac{\sqrt{2} \operatorname{cis}(-\frac{\pi}{6})}{\sqrt{2} \operatorname{cis} \frac{\pi}{4}} = \operatorname{cis}(-\frac{5\pi}{12})$

(iii) Equating real parts of (i) & (ii)
 $\frac{\sqrt{3}-1}{2} = \sqrt{2} \cos(-\frac{5\pi}{12})$
 $\frac{\sqrt{3}-1}{2\sqrt{2}} = \cos(-\frac{5\pi}{12})$
 $\cos$ is EVEN, $\cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$

6(b)(i) $(x+iy)^2 = 24-10i$, x, y real
 $x^2 - y^2 + 2ixy = 24-10i$
 Equating reals, $x^2 - y^2 = 24$ (1)
 Equating imaginaries, $2xy = -10$ (2)
 $y = -\frac{5}{x}$ (2)
 Sub (2) in (1):
 $x^2 - (-\frac{5}{x})^2 = 24$
 $x^2 - \frac{25}{x^2} = 24$
 $x^4 - 24x^2 - 25 = 0$
 $(x^2 - 25)(x^2 + 1) = 0$
 As x is real, $x = \pm 5$
 As $y = -\frac{5}{x}$, $y = \mp 1$
 $\therefore \pm \sqrt{24-10i} = \pm(5-i)$

Q(6)(b)(ii) $z^2 + (1+i)z - 6+3i = 0$ (i)

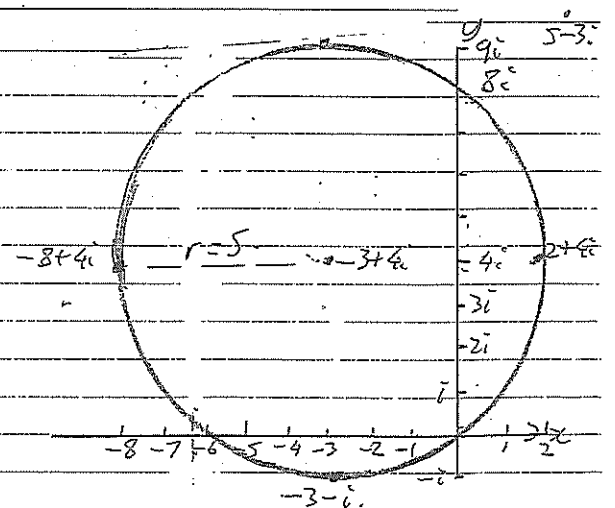
Using $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 $z = \frac{(-1-i) \pm \sqrt{(1+i)^2 - 4 \times 1 \times (-6+3i)}}{2 \times 1}$

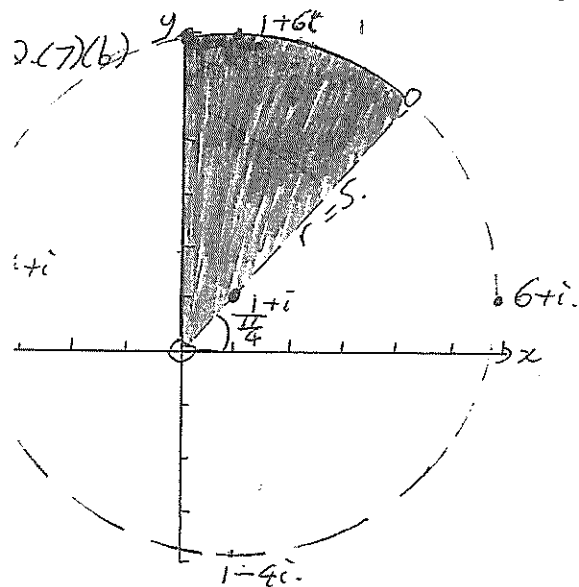
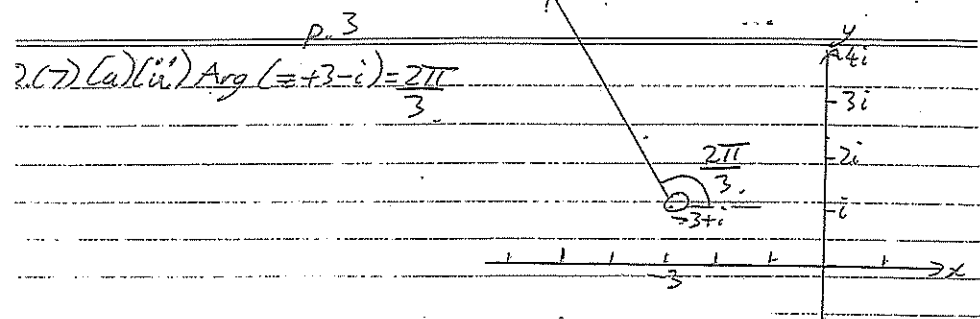
$16 - 16i\sqrt{3} = 32 \operatorname{cis}(-\frac{\pi}{3})$

The five 5th roots of $16 - 16i\sqrt{3}$ are:
 $2 \operatorname{cis}(\frac{-\pi/3 + 2k\pi}{5})$, $k=0,1,2,3,4$
 $= 2 \operatorname{cis} -\frac{\pi}{15}, 2 \operatorname{cis} \frac{\pi}{5}, 2 \operatorname{cis} \frac{11\pi}{15}, 2 \operatorname{cis} \frac{17\pi}{15}, 2 \operatorname{cis} \frac{23\pi}{15}$

Q(7)(a)(i)

$|z + 3 - 4i| = 5$
 Circle centre $-3+4i$
 $r = 5$





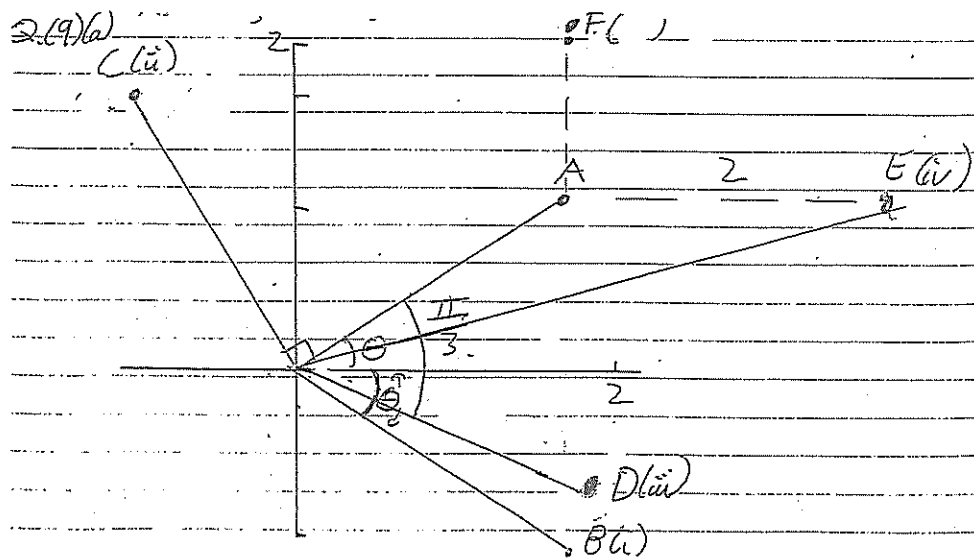
(8) (i) If $z = \cos \theta + i \sin \theta$
 By De Moivre's $z^n = \cos n\theta + i \sin n\theta$
 $= \cos n\theta + i \sin n\theta$
 $= \cos n\theta + i \sin n\theta = \cos n\theta + i \sin n\theta$ (as cos even, sin odd)
 $= 2i \sin n\theta$

(ii) Hence if $z = \cos \theta + i \sin \theta$
 $(z - \bar{z})^5 = (2i \sin \theta)^5$
 $z^5 - 5z^4\bar{z} + 10z^3\bar{z}^2 - 10z^2\bar{z}^3 + 5z\bar{z}^4 - \bar{z}^5$
 $= (z^5 - \bar{z}^5) - 5(z^4\bar{z} - z\bar{z}^4) + 10(z^3\bar{z}^2 - z^2\bar{z}^3)$
 $2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta$
 $-\frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta$
 $= 32i \sin 5\theta = 32i$
 $= \sin 5\theta$

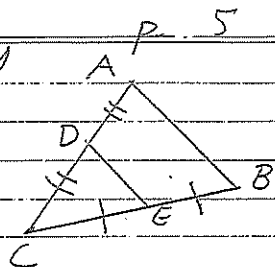
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Q. (8) (i) $\sin^5 \theta = \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta$
 Sub. in $\theta = \frac{\pi}{5}$
 $\sin^5 \left(\frac{\pi}{5}\right) = \frac{1}{16} \sin \pi - \frac{5}{16} \sin \frac{3\pi}{5} + \frac{5}{8} \sin \frac{\pi}{5}$
 As $\sin \pi = 0$ & $\sin \frac{3\pi}{5} = 3 \sin \frac{\pi}{5} - 4 \sin^3 \frac{\pi}{5}$
 $\sin^5 \frac{\pi}{5} = -\frac{5}{16} (3 \sin \frac{\pi}{5} - 4 \sin^3 \frac{\pi}{5}) + \frac{5}{8} \sin \frac{\pi}{5}$
 $= -\frac{5}{16} (3 \sin \frac{\pi}{5} - 4 \sin^3 \frac{\pi}{5}) + \frac{5}{8} \sin \frac{\pi}{5}$
 $= -\frac{5}{16} (3 \sin \frac{\pi}{5} - 4 \sin^3 \frac{\pi}{5}) + \frac{5}{8} \sin \frac{\pi}{5}$
 $= 16 \sin^5 \frac{\pi}{5} - 20 \sin^3 \frac{\pi}{5} + 5 \sin \frac{\pi}{5}$
 As $\sin \frac{\pi}{5} \neq 0$
 $0 = 16 \sin^4 \frac{\pi}{5} - 20 \sin^2 \frac{\pi}{5} + 5$
 $\therefore \sin^2 \frac{\pi}{5}$ is a root of $16x^2 - 20x + 5 = 0$
 $\therefore \sin^2 \frac{\pi}{5} = \frac{20 \pm \sqrt{(-20)^2 - 4 \times 16 \times 5}}{2 \times 16}$

$\sin^2 \frac{\pi}{5} = \frac{5 \pm \sqrt{5}}{8}$
 Noting $\frac{\pi}{5} < \frac{\pi}{4}$, $\sin \frac{\pi}{5} < \sin \frac{\pi}{4}$ i.e. $\sin \frac{\pi}{5} < \frac{1}{\sqrt{2}} \therefore \sin^2 \frac{\pi}{5} < \frac{1}{2}$
 $\therefore \sin^2 \frac{\pi}{5} = \frac{5 - \sqrt{5}}{8} \Rightarrow \sin \frac{\pi}{5} = \sqrt{\frac{5 - \sqrt{5}}{8}}$



Q. (9)(b) ↑ y



$$(i) \vec{DC} = \frac{1}{2} \vec{AC} \quad \vec{EC} = \frac{1}{2} \vec{BC}$$

$$= \frac{1}{2}(c-a) \quad = \frac{1}{2}(c-b)$$

$$(ii) \vec{DE} = \vec{DC} + \vec{CE}$$

$$= \frac{1}{2}(c-a) + \frac{1}{2}(b-c)$$

$$= \frac{1}{2}(b-a)$$

$$\vec{AB} = b-a$$

$$= 2 \times \vec{DE}$$

Q. (10)(a) (i) $\frac{z-2i}{z+4}$ is real

$$\frac{z-2i}{z+4} = \frac{x+(y-2)i}{(x+4)+iy} \times \frac{(x+4)-iy}{(x+4)-iy}$$

$$= \frac{x(x+4) - ixy + i(xy+4y-2x-8) + y(y-2)}{(x+4)^2 + y^2}$$

If $\frac{z-2i}{z+4}$ is real then $xy + 4y - 2x - 8 - xy = 0$

$$x - 2y + 4 = 0$$

(ii) $|z-7-6i| = 2|z+5+3i|$

If $z = x+iy$

$$(x-7)^2 + (y-6)^2 = 4[(x+5)^2 + (y+3)^2]$$

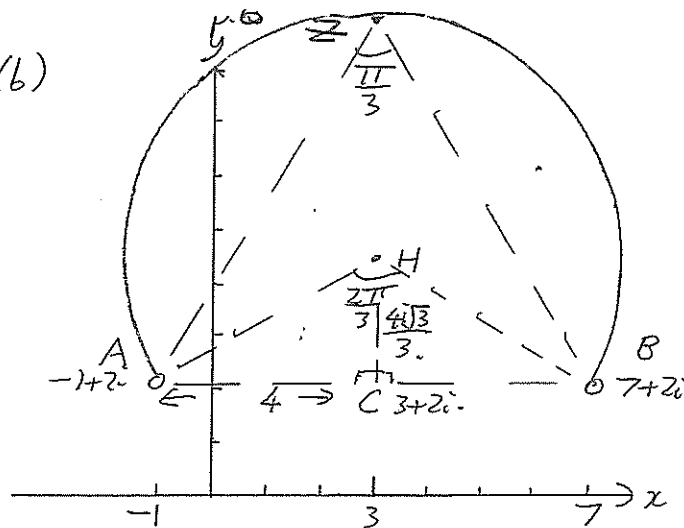
$$x^2 - 14x + 49 + y^2 - 12y + 36 = 4x^2 + 40x + 100 + 4y^2 + 24y + 36$$

$$0 = 3x^2 + 54x + 3y^2 + 36y + 51$$

$$0 = x^2 + 18x + y^2 + 12y + 17$$

$$100 = (x+9)^2 + (y+6)^2$$

Q. (10)(b)



$$\text{Arg} \left[\frac{(z-7-2i)}{(z+1+2i)} \right] = \frac{\pi}{3}$$

Letting $O_{loc} = z$

$= A, 7$

$$\angle A \cong B = \frac{\pi}{3}$$

Circle centre = H.

$$\angle AHB = \frac{2\pi}{3} \quad [\angle \text{at centre} = 2 \times \angle \text{at circumference}]$$

From observation, x at circle centre = 3.

Letting midpoint $AB = C = 3+2i$.

$AH = BH$ (circle radii).

$$\angle HAC = \frac{\pi}{6} \quad [\angle \text{ opposite } = \text{ sides in isosceles } \triangle AHC]$$

$$HAC = 4 + \tan \frac{\pi}{6} = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3} \quad [\text{using trigonometry in } \triangle AHC]$$

$$\therefore H = 3 + \left(2 + \frac{4\sqrt{3}}{3}\right)i$$

$$AH = \frac{4}{\cos \frac{\pi}{6}} = \frac{8}{\sqrt{3}}$$

$\therefore$ Locus is circle with centre $3 + \left(2 + \frac{4\sqrt{3}}{3}\right)i$, $r = \frac{8}{\sqrt{3}}$, where $y > 2$.

$$\therefore \text{Equation is } (x-3)^2 + \left[y - 2 - \frac{4\sqrt{3}}{3}\right]^2 = \frac{64}{3}, y > 2.$$

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Q (ii) (a) (i) $z^9 - 1 = 0$

$z = \text{cis } \frac{2\pi}{9}, k=0, 1, 2, \dots, 8$

(ii) As $1, \omega, \omega^2, \dots, \omega^8$ are roots of $z^9 - 1 = 0$

By $\alpha + \beta + \dots = -b$

$1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 + \omega^7 + \omega^8 = 0$

$\omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 + \omega^7 + \omega^8 = -1$

(iii) $\frac{1}{\omega} = \frac{\omega^8}{\omega^9} = \omega^8 = \omega^8$

(iv) Similarly, $\frac{1}{\omega^2} = \omega^7, \frac{1}{\omega^3} = \omega^6$ & $\frac{1}{\omega^4} = \omega^5$

Hence $\omega + \omega^2 + \omega^3 + \omega^4 + \frac{1}{\omega} + \frac{1}{\omega^2} + \frac{1}{\omega^3} + \frac{1}{\omega^4}$
 $= \omega + \omega^2 + \omega^3 + \omega^4 + \omega^8 + \omega^7 + \omega^6 + \omega^5$
 $= -1$ [by (ii)]

(v) As $\omega + \frac{1}{\omega} + \omega^2 + \frac{1}{\omega^2} + \omega^3 + \frac{1}{\omega^3} + \omega^4 + \frac{1}{\omega^4} = -1$

Using $\omega + \frac{1}{\omega} = 2\cos\frac{2\pi}{9}, \omega^2 + \frac{1}{\omega^2} = 2\cos\frac{4\pi}{9}$ etc.

$2\cos\frac{2\pi}{9} + 2\cos\frac{4\pi}{9} + 2\cos\frac{6\pi}{9} + 2\cos\frac{8\pi}{9} = -1$

Noting $\cos\frac{6\pi}{9} = -\frac{1}{2}$ & $\cos\frac{8\pi}{9} = -\cos\frac{\pi}{9}$

$2\cos\frac{2\pi}{9} + 2\cos\frac{4\pi}{9} - 1 - 2\cos\frac{\pi}{9} = -1$

$2\cos\frac{2\pi}{9} + 2\cos\frac{4\pi}{9} - 1 - 2\cos\frac{\pi}{9} = -1$

$\cos\frac{2\pi}{9} + \cos\frac{4\pi}{9} = \cos\frac{\pi}{9}$

b) (i) $z^9 - 1 = (z-1)(z^8 + z^7 + \dots + 1)$ (by $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$)
 $= (z-1)(z - \text{cis}\frac{2\pi}{9})(z - \text{cis}\frac{4\pi}{9})(z^6 + z^5 + \dots + 1)$

Using $(z - \text{cis}\frac{2\pi}{9})(z - \text{cis}\frac{4\pi}{9}) = (z - \text{cis}\frac{2\pi}{9})(z - \text{cis}(-\frac{2\pi}{9}))$
 $= (z^2 - z(\cos\frac{2\pi}{9} + i\sin\frac{2\pi}{9} + \cos\frac{2\pi}{9} - i\sin\frac{2\pi}{9}) + 1) = z^2 - 2z\cos\frac{2\pi}{9} + 1$
 $z^9 - 1 = (z-1)(z^2 + z + 1)(z^6 + z^5 + \dots + 1)$ (1)

$z^9 - 1$ also $= (z-1)(z - \text{cis}\frac{2\pi}{3})(z - \text{cis}\frac{4\pi}{3})(z - \text{cis}\frac{2\pi}{9})(z - \text{cis}\frac{10\pi}{9})$
 $(z - \text{cis}\frac{4\pi}{9})(z - \text{cis}\frac{14\pi}{9})(z - \text{cis}\frac{8\pi}{9})(z - \text{cis}\frac{16\pi}{9})$
 $= (z-1)(z^2 + z + 1)(z - \text{cis}\frac{2\pi}{9})(z - \text{cis}\frac{16\pi}{9})(z - \text{cis}\frac{4\pi}{9})(z - \text{cis}\frac{14\pi}{9})$
 $(z - \text{cis}\frac{8\pi}{9})(z - \text{cis}\frac{10\pi}{9})$

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(ii) (b) (i) (continued):

Noting $(z - \text{cis}\frac{2\pi}{9})(z - \text{cis}\frac{16\pi}{9})$
 $= (z - \text{cis}\frac{2\pi}{9})(z - \text{cis}(-\frac{2\pi}{9}))$
 $= z^2 - z(\cos\frac{2\pi}{9} + i\sin\frac{2\pi}{9} + \cos\frac{2\pi}{9} - i\sin\frac{2\pi}{9}) + 1$ [as cos even, sin odd]
 $= z^2 - 2z\cos\frac{2\pi}{9} + 1$
 [Note: You ARE allowed to assume in Q.11 that $\text{cis } n\theta + \text{cis}(-n\theta) = 2\cos n\theta$.

Similarly, $(z - \text{cis}\frac{4\pi}{9})(z - \text{cis}\frac{14\pi}{9}) = z^2 - 2z\cos\frac{4\pi}{9} + 1$
 & $(z - \text{cis}\frac{8\pi}{9})(z - \text{cis}\frac{10\pi}{9}) = z^2 - 2z\cos\frac{8\pi}{9} + 1$
 $= z^2 + 2z\cos\frac{\pi}{9} + 1$

Hence $z^9 - 1$ also $=$

$(z-1)(z^2 + z + 1)(z^2 - 2z\cos\frac{2\pi}{9} + 1)(z^2 - 2z\cos\frac{4\pi}{9} + 1)(z^2 + 2z\cos\frac{\pi}{9} + 1)$

Equating (1) & (2) & dividing BS by $(z-1)(z^2 + z + 1)$
 $z^6 + z^3 + 1 = (z^2 - 2z\cos\frac{2\pi}{9} + 1)(z^2 - 2z\cos\frac{4\pi}{9} + 1)(z^2 + 2z\cos\frac{\pi}{9} + 1)$

(ii) Hence, dividing (i) by z^3

$\frac{z^3 + 1 + \frac{1}{z^3}}{z^3} = (z + \frac{1}{z} - 2\cos\frac{2\pi}{9})(z + \frac{1}{z} - 2\cos\frac{4\pi}{9})(z + \frac{1}{z} + 2\cos\frac{\pi}{9})$

Letting $z = \text{cis } \theta, z + \frac{1}{z} = 2\cos\theta$

$2\cos 3\theta + 1 = (2\cos\theta - 2\cos\frac{2\pi}{9})(2\cos\theta - 2\cos\frac{4\pi}{9})(2\cos\theta + 2\cos\frac{\pi}{9})$

(iii) Hence letting $\theta = \frac{\pi}{9}$, so $\cos 3\theta = \cos\theta = 0$

$1 = -2\cos\frac{2\pi}{9} \times -2\cos\frac{4\pi}{9} \times 2\cos\frac{\pi}{9}$

$\div 8$

$\frac{1}{8} = \cos\frac{\pi}{9} \cos\frac{2\pi}{9} \cos\frac{4\pi}{9}$

[or $\cos\frac{\pi}{9} \cos\frac{2\pi}{9} \cos\frac{4\pi}{9} = \frac{1}{8}$]

End of solutions!!